Identifying Treatment Effects on Categorical Outcomes in IV Models

Onil Boussim
Department of Economics, Pennsylvania State University,
University Park, PA 16802
oib5044@psu.edu

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Abstract

I develop a nonparametric framework for identifying treatment effects on unordered categorical outcomes using a binary instrumental variable. A local copula decomposition expresses each observed joint probability of outcome and treatment as a function of marginal potential-outcome probabilities and a Gaussian copula parameter encoding selection on unobservables. With a binary instrument, this system is underidentified by two degrees of freedom per outcome category. Under *association similarity*, a new assumption requiring the copula parameter to be invariant across treatment states, the system becomes exactly identified and the full distribution of potential outcomes is nonparametrically point-identified. Two weaker alternatives, *monotonic association* and *bounded association*, deliver sharp partial identification bounds and a formal sensitivity analysis. I introduce a semiparametric IV multinomial logit model estimated by two-step maximum likelihood, prove $\sqrt{n}$-consistency, and conduct inference via bootstrap. Monte Carlo simulations confirm low bias and accurate coverage. An application to the Democratic Republic of Congo shows that improved market access reduces primary-sector employment by 21 percentage points and raises industrial employment by 14 percentage points.

Key Words: Association similarity; Categorical outcomes; Copula; Instrumental variables; Partial identification; Treatment effects

1. INTRODUCTION

A central challenge in causal inference is endogeneity: many treatments of interest result from choices made by agents who act on private information that also affects the outcomes the researcher seeks to explain. Individuals decide whether to enroll in education programs, firms choose whether to adopt new technologies, and governments design tax or subsidy schemes. These decisions are typically correlated with unobserved factors, motivation, ability, or local market conditions, that simultaneously influence outcomes of interest. Instrumental variables (IV) methods provide classical solutions by leveraging exogenous variation in the probability of treatment to disentangle causal effects from endogenous selection.

In practice, researchers confronted with an unordered categorical outcome, such as employment sector, disease subtype, or product choice, typically proceed in one of two ways. The most common approach is to convert the outcome into a binary indicator (e.g., employed versus not, formal versus informal sector) and apply standard binary-outcome IV methods. While tractable, this strategy discards the distributional content of the original outcome: it cannot reveal whether treatment shifts the entire composition of choices, whether gains in one category comes at the expense of another, or how effects vary across categories. A second approach is to impose a parametric model, such as a multinomial logit or probit and introduce endogeneity corrections via correlated random effects or control function terms. This strategy recovers richer distributional information but at the cost of strong distributional assumptions on the unobservables, whose misspecification can severely bias estimates.

Both approaches share a common limitation: they fail to exploit, in a transparent nonparametric fashion, the identifying power of the instrument jointly across all outcome categories. With a categorical outcome having q categories and a single binary instrument, the number of observable moments increases proportionally with q, but so does the dimension of the unknown objects of interest. The identification challenge is therefore not merely one of functional form: it is a question of whether, and under what restrictions on the dependence structure between unobservables and treatment selection, the data identify the full distribution of potential outcomes.

I develop a flexible nonparametric framework that directly confronts this challenge. I work with a multinomial log-odds parametrization of the categorical potential outcome distributions and apply a local copula decomposition to express each observed joint probability of outcome and treatment as a function of three objects: (i) the marginal probability of the relevant potential outcome category, (ii) the marginal probability of treatment under the given instrument value, and (iii) a Gaussian copula parameter that captures the dependence between the latent indicator of category membership and the latent treatment indicator. The first two objects are recoverable from the potential outcome distributions and the first stage; the third encodes the selection mechanism.

This decomposition, the *local copula decomposition*, reveals precisely why identification is hard in this setting and what additional structure can resolve it. With a binary instrument and binary treatment, there are four observable moments per non-baseline category but six unknown parameters: two log-odds parameters (one per potential outcome state) and four copula parameters (one for each combination of treatment state and instrument value). Without further restrictions, the model is underidentified by two degrees of freedom per category.

My main identification result rests on *association similarity*: the copula parameter governing the dependence between latent category membership and the treatment indicator is the same whether one looks at the treated or the untreated potential outcome. This invariance of the selection mechanism across potential states reduces the number of unknowns per category from six to four, exactly matching the number of observable moments and delivering point identification. The result holds for any number of categories and requires no parametric assumption on the distribution of unobservables.

To illustrate, consider an encouragement experiment for a job training program where the outcome is employment in one of three sectors: manufacturing, services, or unemployment. Unobserved traits such as motivation affect both treatment take-up and sectoral outcomes. Association similarity requires that the correlation between these traits and sectoral employment tendencies is the same whether the individual receives training or not. The program may shift the entire distribution of employment outcomes, but the underlying correlation between motivation and employment tendency remains stable across treatment

states. Under this assumption, one can cancel out the selection bias by exploiting how the instrument shifts treatment probabilities, yielding nonparametric formulas for the full potential outcome distribution.

I complement the point identification result with two partial identification alternatives. *Monotonic association* permits the copula parameter to differ across treatment states but restricts the direction of the difference. *Bounded association* allows arbitrary deviations but constrains their magnitude by a user-specified sensitivity parameter δ. Both assumptions deliver sharp identified sets that collapse to the point-identified value when $\delta = 0$, providing a transparent sensitivity analysis.

Building on the identification result, I introduce a semiparametric IV multinomial logit model and a two-step estimator. In the first step, I estimate the treatment selection equation by logistic regression. In the second step, I maximize a multinomial log-likelihood incorporating the copula structure, using the first-step estimates as plug-ins. The estimator is $\sqrt{n}$-consistent and asymptotically normal, and inference is conducted by nonparametric bootstrap.

Monte Carlo simulations confirm that the estimator is nearly unbiased and that the bootstrap delivers reliable confidence intervals at sample sizes typical of survey data. I apply the framework to the Democratic Republic of Congo, using local topography as an instrument for market access. Improved market access induces a substantial reallocation of labor: the probability of primary-sector employment falls by 21.4 percentage points, while industrial employment rises by 13.6 percentage points.

Contributions and Outline

The paper makes several methodological contributions. First, it extends IV identification theory to unordered categorical outcomes without parametric assumptions, filling a gap relative to the ordered-outcome literature of Chernozhukov et al. [2015]. Second, it introduces association similarity and the local copula decomposition as a transparent organizing framework. Third, it provides sharp partial identification results and formal sensitivity analysis. Fourth, it delivers a tractable two-step estimator with bootstrap inference.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3

introduces the analytical framework and the main identification result. Section 4 develops the partial identification results. Section 5 presents the estimation strategy. Section 6 reports Monte Carlo evidence. Section 7 contains the empirical application, and Section 8 concludes. Proofs are collected in Appendix A.

2. RELATED LITERATURE

The identification of causal effects with categorical outcomes poses distinct challenges compared to the scalar case. While IV methods for continuous outcomes are now well established (see the survey by Mogstad and Torgovitsky [2024]), progress for discrete outcomes has been slower.

Parametric approaches. Early contributions focused on parametric models, such as multinomial probit and logit models with correlated errors (van de Ven and van Praag 1981; Dubin and McFadden 1984; Dubin and Rivers 1989), which achieve identification only under strong distributional assumptions. Related work on bivariate probit models includes Freedman and Sekhon [2010], Mourifie [2015], Han [2021], Han and Lee [2019], and Acerenza [2023], all of which remain largely parametric. The present paper departs from this strand by avoiding distributional assumptions on the unobservables entirely.

Nonparametric point identification. Angrist et al. [1996] provided an early nonparametric contribution by showing the identification of local average treatment effects in IV models with categorical outcomes; however, this delivers the potential outcome distribution only for compliers rather than for the full population. More recently, Chernozhukov et al. [2015] developed a local Gaussian representation (LGR) that, under a copula invariance assumption, enables nonparametric identification in selection models. Their strategy, however, is restricted to *ordered* categorical outcomes, since it exploits cumulative distribution functions. I adapt and extend the LGR logic to unordered categorical outcomes by working with the multinomial log-odds parametrization. Using category-specific marginals rather than cumulative probabilities allows the framework to handle genuinely unordered outcomes in which no natural ordering exists.

Partial identification. Manski [1990] derived the sharpest bounds on treatment effects consistent with the data and basic assumptions. Balke and Pearl [1997] obtained sharp IV bounds for binary outcomes under no-defiers. Vytlacil and Yildiz [2007] and Shaikh and Vytlacil [2011] extended partial identification results to more general limited dependent variable settings. Beresteanu and Molinari [2008] and Chesher [2010] study sharp identification regions using random set theory and quantile methods, respectively. This paper contributes by providing sharp bounds under two novel weakened forms of association similarity. The framework also relates to the sensitivity analysis literature of Rosenbaum [1987] and, in the IV context, Conley et al. [2012] and Nevo and Rosen [2012]: bounded association serves as a formal sensitivity parameter.

Estimation. Newey [1994] provides the general framework for semiparametric two-step estimators underlying the estimation strategy in Section 5. Horowitz [2019] provides foundations for the bootstrap inference procedure used in the empirical application.

3. FRAMEWORK AND POINT IDENTIFICATION

3.1 Analytical Framework

I adopt the potential outcomes framework. Let the treatment be binary, $D \in \{0, 1\}$, and let the instrument also be binary, $Z \in \{0, 1\}$. The outcome of interest, Y, takes values in a finite set of q mutually exclusive and exhaustive unordered categories:

$$Y \in \mathcal{C} = \{c_1, c_2, \ldots, c_q\}.$$

Because the outcome is unordered, cumulative distribution functions and ordered-threshold models are not applicable; one must work directly with the vector of category probabilities.

For each individual, define potential outcomes Y_0 and Y_1, where Y_d denotes the outcome that would be observed under treatment status $D = d$. The objects of interest are the full distributions of potential outcomes: for each $d \in \{0, 1\}$,

$$\pi_d = \big(\mathbb{P}(Y_d = c_1),\, \ldots,\, \mathbb{P}(Y_d = c_q)\big)^\top \in \mathcal{S}^{q-1},$$

where $\mathcal{S}^{q-1}$ is the $(q-1)$-dimensional probability simplex. Category-specific average treatment effects are $\mathbb{P}(Y_1 = c_k) - \mathbb{P}(Y_0 = c_k)$ for each k.

I introduce potential treatment indicators $D_z \in \{0,1\}$, where D_z is the treatment decision under $Z = z$. The observed variables satisfy

$$D = ZD_1 + (1-Z)D_0, \qquad Y = DY_1 + (1-D)Y_0.$$

Treatment selection is modeled through latent indices: $D_z = \mathbf{1}\{U_z \geq 0\}$ for latent variables U_0, U_1 capturing unobserved heterogeneity. I impose the two standard IV assumptions.

Assumption 1 (Independence)**.** $(Y_0, Y_1, U_0, U_1) \perp\!\!\!\perp Z$.

Assumption 1 is the exclusion restriction: the instrument is statistically independent of all potential outcomes and latent treatment indices. It rules out direct effects of Z on Y_d, unobserved confounders jointly affecting Z and (Y_0, Y_1), and self-selection into instrument values correlated with potential treatment decisions.

Assumption 2 (Relevance)**.** $p_1 \neq p_0$, *where* $p_z = \mathbb{P}(D = 1 \mid Z = z)$.

Both p_0 and p_1 are directly identified from the data.

3.2 The Local Copula Decomposition

I work with the multinomial parametrization for $Y_d \in \{c_1, \ldots, c_q\}$, taking c_q as the baseline. Define the log-odds parameters

$$\mu_k^d = \log\left(\frac{\mathbb{P}(Y_d = c_k)}{\mathbb{P}(Y_d = c_q)}\right), \quad k = 1, \ldots, q-1,$$

with $\mu_q^d = 0$ by convention. Under the softmax parametrization,

$$\pi_k^d = \mathbb{P}(Y_d = c_k) = \frac{e^{\mu_k^d}}{1 + \sum_{j=1}^{q-1} e^{\mu_j^d}}, \quad k = 1, \ldots, q-1.$$

The map $\ell(\pi^d) = (\mu_1^d, \ldots, \mu_{q-1}^d)$ is a smooth diffeomorphism from $\mathcal{S}^{q-1}$ to $\mathbb{R}^{q-1}$.

I use the standard bivariate Gaussian copula

$$\Lambda_2(u, v, \omega) = \Phi_2\big(\Phi^{-1}(u),\, \Phi^{-1}(v);\, \omega\big), \quad (u, v, \omega) \in (0,1)^2 \times (-1, 1),$$

where $\Phi_2(\cdot, \cdot; \omega)$ is the standard bivariate normal CDF with correlation ω and Φ is the standard normal CDF. The map $\omega \mapsto \Lambda_2(u, v, \omega)$ is strictly increasing on $(-1, 1)$ for any fixed $(u, v) \in (0,1)^2$.

Lemma 3.1 (Local Copula Decomposition)**.** *Under Assumptions 1 and 2, for each* $z \in \{0,1\}$ *and* $k = 1, \ldots, q-1$*,*

$$\mathbb{P}(Y = c_k, D = 1 \mid Z = z) = \Lambda_2\left(\frac{e^{\mu_k^1}}{1 + \sum_{j=1}^{q-1} e^{\mu_j^1}},\ p_z,\ \omega_{1,k,z}\right), \tag{3.1}$$

$$\mathbb{P}(Y = c_k, D = 0 \mid Z = z) = \Lambda_2\left(\frac{e^{\mu_k^0}}{1 + \sum_{j=1}^{q-1} e^{\mu_j^0}},\ 1 - p_z,\ -\omega_{0,k,z}\right), \tag{3.2}$$

where $\omega_{d,k,z} \in (-1,1)$ *is the copula parameter governing the dependence between* $\mathbf{1}\{Y_d = c_k\}$ *and* D_z*, and satisfies* $\operatorname{sign}(\omega_{d,k,z}) = \operatorname{sign}\big(\operatorname{Cov}(\mathbf{1}\{Y_d = c_k\},\ D_z)\big)$*.*

The proof is in Appendix A. When $\omega_{d,k,z} = 0$, the potential outcome Y_d and the treatment indicator D_z are independent given $Z = z$, the local random assignment case.

Remark 1. The representation in Lemma 3.1 parallels the local Gaussian representation of Chernozhukov et al. [2015]. The key difference is that their LGR is formulated for ordered outcomes via the conditional CDF, while the present representation uses the multinomial log-odds parametrization, which requires no ordering among categories.

3.3 The Identification Problem

Lemma 3.1 reveals exactly why identification is problematic without further restrictions. For each non-baseline category c_k ($k = 1, \ldots, q-1$), there are four observable moment conditions:

$$\big\{\mathbb{P}(Y = c_k, D = d \mid Z = z) : d \in \{0,1\},\ z \in \{0,1\}\big\}.$$

However, these four moments depend on six unknown scalar parameters: μ_k^0, μ_k^1, $\omega_{0,k,0}$, $\omega_{0,k,1}$, $\omega_{1,k,0}$, and $\omega_{1,k,1}$. Since $4 < 6$, the system is underidentified by two degrees of freedom per category. Absent further restrictions, the identified set reduces to wide Manski bounds [Manski, 1990]. Point identification therefore requires two restrictions on the four copula parameters $\{\omega_{d,k,z}\}$.

3.4 Point Identification Under Association Similarity

Assumption 3 (Association Similarity)**.** *For all* $z \in \{0,1\}$ *and* $k = 1, \ldots, q-1$*:*

$$\omega_{0,k,z} = \omega_{1,k,z} \equiv \omega_{k,z}.$$

Assumption 3 states that the copula parameter governing the association between the latent category-k indicator and the treatment indicator D_z is the same under both potential outcome states $d = 0$ and $d = 1$. Three features deserve emphasis.

First, the assumption does not require treatment to have no effect. It permits treatment to shift the marginal distribution of outcomes (changing $\mathbb{P}(Y_1 = c_k)$ relative to $\mathbb{P}(Y_0 = c_k)$) while requiring that the underlying correlation between latent heterogeneity and outcome tendencies remains stable across treatment states. The log-odds parameters μ_k^0 and μ_k^1 are free to differ; only the copula parameters are equated.

Second, the assumption restricts the dependence structure, not the potential outcomes themselves. Treatment may reshape the entire distribution across categories, but the correlation between the latent propensity to be in category k and the propensity to take treatment is the same in both potential states.

Third, the assumption is compatible with a broad class of latent index models. Consider

$$Y_d = h_d(X, \varepsilon), \quad D_z = \mathbf{1}\{g_z(X, \eta) \geq 0\},$$

where h_d and g_z are arbitrary functions. If the joint distribution of (ε, η) does not depend on d, then Assumption 3 holds. This parallels the notion of rank similarity in the continuous potential outcomes literature (Chernozhukov and Hansen 2005). Under Assumption 3, the number of unknowns per category reduces from six to four: $(\mu_k^0, \mu_k^1, \omega_{k,0}, \omega_{k,1})$. The system is now exactly identified, four equations and four unknowns per category. To state the identification result precisely, define

$$e_k(d, z) = \frac{\Phi^{-1}(p_z) - \omega_{k,z}\,\Phi^{-1}\big(\pi_k^d\big)}{\sqrt{1 - \omega_{k,z}^2}},$$

and the inverse Mills ratio-type quantities

$$\lambda_k(d, z) \equiv \frac{\phi\big(e_k(d, z)\big)}{\Phi\big(e_k(d, z)\big)}, \qquad \bar{\lambda}_k(d, z) \equiv \frac{\phi\big(e_k(d, z)\big)}{\Phi\big(-e_k(d, z)\big)}.$$

Assumption 4 (Rank Condition)**.** *For each $k = 1, \ldots, q - 1$:*

$$\frac{\lambda_k(1, 1)}{\lambda_k(1, 0)} \neq \frac{\bar{\lambda}_k(0, 1)}{\bar{\lambda}_k(0, 0)}.$$

Assumption 4 requires that the ratio of selection correction terms across instrument values differs between the treated and untreated potential outcome states. It rules out a knife-edge case in which the moment mapping is not locally injective and holds generically.

Proposition 3.2 (Point Identification)**.** *Suppose Assumptions 1, 2, 3, and 4 hold, and that* $|\omega_{k,z}| < 1$ *for all* k *and* z*. Then the parameter vector*

$$\theta = \left((\mu_k^0,\, \mu_k^1,\, \omega_{k,0},\, \omega_{k,1})_{k=1}^{q-1}\right)$$

is point-identified as the unique solution to the nonlinear system formed by equations (3.1)–(3.2) with $\omega_{0,k,z} = \omega_{1,k,z}$*. In particular, the potential outcome distributions* π_0 *and* π_1 *are point-identified.*

The proof is in Appendix A. The system decomposes into $(q-1)$ independent four-dimensional subsystems, one per non-baseline category. Each subsystem reduces to a one-dimensional fixed-point problem. Assumption 4 ensures that the fixed-point map has derivative bounded away from one everywhere on the feasible interval, so the map is either a contraction or has a contracting inverse. In either case, at most one fixed point exists, establishing global injectivity of the moment mapping.

Remark 2 (Conditional identification). In empirical practice, Assumption 3 may be more plausible after conditioning on observed covariates X. All probabilities and dependence parameters may then be interpreted as conditional on X, with identification proceeding within strata defined by X and marginal potential outcome distributions recovered by integrating over the empirical distribution of X.

4. PARTIAL IDENTIFICATION UNDER WEAKER ASSUMPTIONS

Assumption 3 imposes exact equality of copula parameters across treatment states. This section introduces two strictly weaker alternatives.

4.1 Monotonic Association

Assumption 5 (Monotonic Association)**.** *For all* $z \in \{0, 1\}$ *and* $k = 1, \ldots, q-1$*:* $\omega_{1,k,z} \geq \omega_{0,k,z}$*.*

Assumption 5 states that the strength of selection on unobservables is weakly larger for the treated potential outcome than for the untreated potential outcome. In the job training example, this says that if motivated workers sort into certain sectors when untrained, they are at least as strongly sorted when trained.

Proposition 4.1 (Bounds Under Monotonic Association)**.** *Under Assumptions 1, 2, and 5, the potential outcome probabilities* (π_k^0, π_k^1) *lie in a sharp identified set* $\mathcal{B}_k^{\text{mono}}$ *characterized by*

$$\mathcal{B}_k^{\text{mono}} = \left\{(\pi_k^0, \pi_k^1, \omega_{0,k,z}, \omega_{1,k,z}) : \textit{(3.1)–(3.2) hold},\ \omega_{1,k,z} \geq \omega_{0,k,z},\ |\omega_{d,k,z}| < 1\right\}.$$

This set is sharp, strictly contains the singleton identified under Assumption 3, and collapses to that singleton when $\omega_{1,k,z} = \omega_{0,k,z}$.

4.2 Bounded Association

Assumption 6 (Bounded Association)**.** *Let* $\delta \geq 0$ *be a user-specified sensitivity parameter. For all* $z \in \{0,1\}$, $d \in \{0,1\}$, *and* $k = 1, \ldots, q-1$: $|\omega_{1,k,z} - \omega_{0,k,z}| \leq \delta$.

When $\delta = 0$, Assumption 6 reduces to Assumption 3 and delivers point identification. As δ increases, the identified set expands and the sensitivity of conclusions to departures from association similarity can be traced out explicitly.

Proposition 4.2 (Bounds Under Bounded Association)**.** *Under Assumptions 1, 2, and 6 with sensitivity parameter* $\delta \geq 0$, *the potential outcome probabilities* (π_k^0, π_k^1) *lie in a sharp identified set* $\mathcal{B}_k^\delta$ *satisfying:*

(i) When $\delta = 0$: $\mathcal{B}_k^0$ *is the singleton given by Proposition 3.2.*

(ii) For $\delta > 0$: $\mathcal{B}_k^\delta$ *is compact and connected, and expands continuously in* δ.

(iii) As $\delta \to \infty$: $\mathcal{B}_k^\delta$ *converges to the Manski [1990] bounds.*

The proofs of Propositions 4.1 and 4.2 are in Appendix A. Proposition 4.2 facilitates a natural sensitivity analysis: the researcher can display how the identified set for any causal estimand expands as δ increases from zero, making transparent the robustness of conclusions to violations of association similarity. The monotone expansion property ensures that

the point-identified estimate is always the most informative statement consistent with the data under this class of assumptions.

5. ESTIMATION

5.1 Nonparametric Estimation

Under the nonparametric specification, the observable moments $m_{k,z}^{(d)} = \mathbb{P}(Y = c_k, D = d \mid Z = z)$ and $p_z = \mathbb{P}(D = 1 \mid Z = z)$ are consistently estimated by their sample analogs:

$$\hat{m}_{k,z}^{(d)} = \frac{\sum_{i=1}^{n} \mathbf{1}\{Y_i = c_k,\ D_i = d,\ Z_i = z\}}{\sum_{i=1}^{n} \mathbf{1}\{Z_i = z\}}, \qquad \hat{p}_z = \frac{\sum_{i=1}^{n} \mathbf{1}\{D_i = 1,\ Z_i = z\}}{\sum_{i=1}^{n} \mathbf{1}\{Z_i = z\}}.$$

The point-identified parameter $\hat{\theta}$ is obtained by solving the nonlinear system

$$\hat{m}_{k,z}^{(1)} = \Lambda_2\left(\frac{\exp(\hat{\mu}_k^1)}{1 + \sum_{j=1}^{q-1} \exp(\hat{\mu}_j^1)},\ \hat{p}_z,\ \hat{\omega}_{k,z}\right), \tag{5.1}$$

$$\hat{m}_{k,z}^{(0)} = \Lambda_2\left(\frac{\exp(\hat{\mu}_k^0)}{1 + \sum_{j=1}^{q-1} \exp(\hat{\mu}_j^0)},\ 1 - \hat{p}_z,\ -\hat{\omega}_{k,z}\right), \tag{5.2}$$

for all $k = 1, \ldots, q-1$ and $z \in \{0, 1\}$. By Proposition 3.2, this system has a unique solution, obtainable by standard nonlinear solvers.

5.2 Semiparametric IV Multinomial Logit Model

When covariates X are available, I propose a semiparametric IV multinomial logit model. Let $W = (X, Z)$ and specify:

$$\mu_k^d(X) = X'\beta_k^d, \tag{5.3}$$

$$\mathbb{P}(D = 1 \mid W) = \Lambda_1(W'\delta_0), \tag{5.4}$$

$$\omega_k(W) = \tanh(W'\gamma_k), \tag{5.5}$$

where $\Lambda_1(\cdot)$ is the logistic CDF and $\tanh(\cdot)$ ensures the copula parameter lies in $(-1, 1)$. The implied conditional probabilities of the potential outcomes follow the multinomial logit form:

$$\pi_k^d(X) = \frac{\exp(X'\beta_k^d)}{1 + \sum_{j=1}^{q-1} \exp(X'\beta_j^d)}, \quad k = 1, \ldots, q-1.$$

The marginal probability of observing category c_k given $W = w$ is

$$\mathbb{P}(Y = c_k \mid W = w) = \Lambda_2\big(\pi_k^1(x),\, \Lambda_1(w'\delta_0),\, \tanh(w'\gamma_k)\big) \\ + \Lambda_2\big(\pi_k^0(x),\, 1 - \Lambda_1(w'\delta_0),\, -\tanh(w'\gamma_k)\big).$$

Algorithm 1 (Two-Step Estimator)**.**

Step 1 (Selection equation). *Estimate δ_0 by logistic regression of D on W:*

$$\hat{\delta}_0 = \arg\min_p \left\{ -\sum_{i=1}^n \big[D_i \log \Lambda_1(W_i'p) + (1 - D_i)\log(1 - \Lambda_1(W_i'p))\big] \right\}.$$

Step 2 (Outcome equation). *Define $J_i(k) = \mathbf{1}\{Y_i = c_k\}$. Estimate $\theta_k = (\beta_k^1, \beta_k^0, \gamma_k)$, $k = 1, \ldots, q-1$, jointly by maximizing the multinomial log-likelihood conditional on $\hat{\delta}_0$:*

$$(\hat{\theta}_k)_{k=1}^{q-1} = \arg\min \Bigg\{ -\sum_{i=1}^n \sum_{k=1}^{q-1} J_i(k) \log\Big(\Lambda_2\big(\pi_k^1(X_i; b_k^1),\, \Lambda_1(W_i'\hat{\delta}_0),\, \tanh(W_i'g_k)\big) \\ + \Lambda_2\big(\pi_k^0(X_i; b_k^0),\, 1 - \Lambda_1(W_i'\hat{\delta}_0),\, -\tanh(W_i'g_k)\big)\Big) \\ + J_i(q) \log\Big(1 - \sum_{k=1}^{q-1} \big[\Lambda_2(\cdots) + \Lambda_2(\cdots)\big]\Big) \Bigg\}.$$

Step 1 is standard logistic regression. Step 2 is a smooth nonlinear optimization problem solved with Newton or quasi-Newton methods, using Gauss-Legendre quadrature to evaluate the bivariate Gaussian copula. The joint multinomial objective enforces the adding-up constraint $\sum_{k=1}^q \pi_k^d = 1$ exactly.

The marginal potential outcome probabilities are estimated by averaging:

$$\hat{\mathbb{P}}(Y_d = c_k) = \frac{1}{n} \sum_{i=1}^n \frac{\exp(X_i'\hat{\beta}_k^d)}{1 + \sum_{j=1}^{q-1} \exp(X_i'\hat{\beta}_j^d)}.$$

Inference is conducted using the nonparametric bootstrap.

Remark 3 (Asymptotic theory)*.* Under standard regularity conditions for two-step M-estimators (Newey 1994), the estimators $\hat{\delta}_0$ and $(\hat{\theta}_k)_{k=1}^{q-1}$ are $\sqrt{n}$-consistent and asymptotically normal. The first-step estimation error propagates into the second step through the copula arguments, and the nonparametric bootstrap automatically accounts for this propagation.

6. MONTE CARLO EVIDENCE

To assess finite-sample performance, I conduct a Monte Carlo study. The data-generating process satisfies association similarity by construction, so the exercise evaluates bias, variance, and coverage of the point-identified estimator.

6.1 Data-Generating Process

The sample size is $n = 3{,}000$ and all results are based on $R = 300$ Monte Carlo replications. For each replication, I draw independently

$$X_i \sim \mathcal{N}(0,1), \qquad Z_i \sim \text{Bernoulli}(0.5),$$

and generate correlated unobservables

$$\varepsilon_i \sim \mathcal{N}(0,1), \qquad \eta_i = 0.4\,\varepsilon_i + \sqrt{0.84}\,\tilde{\eta}_i, \quad \tilde{\eta}_i \sim \mathcal{N}(0,1) \text{ independent of } \varepsilon_i,$$

so the degree of endogeneity is $\omega = 0.4$. Treatment is determined by

$$D_i = \mathbf{1}\{0.3\,X_i + 0.8\,Z_i + \eta_i \geq 0\},$$

yielding a strong first stage with $\hat{p}_1 - \hat{p}_0 \approx 0.32$ in every replication.

The outcome is binary, with categories $\mathcal{C} = \{\textit{good},\ \textit{bad}\}$ where *bad* serves as the baseline. Potential outcome probabilities are generated by

$$\text{logit}\,\mathbb{P}(Y_{1i} = \textit{good} \mid X_i, \varepsilon_i) = 1.0 + 0.5\,X_i + 0.6\,\varepsilon_i, \tag{6.1}$$

$$\text{logit}\,\mathbb{P}(Y_{0i} = \textit{good} \mid X_i, \varepsilon_i) = 0.2 + 0.5\,X_i + 0.6\,\varepsilon_i. \tag{6.2}$$

The common factor $0.6\,\varepsilon_i$ ensures association similarity holds exactly. The observed outcome is $Y_i = D_i\,Y_{1i} + (1 - D_i)\,Y_{0i}$.

Integrating over the marginal distribution of (X, ε) yields the population targets:

$$\pi^1_{good} \approx 0.708, \qquad \pi^0_{good} \approx 0.544, \qquad \text{ATE} \approx 0.164.$$

6.2 Results

The estimator is the nonparametric two-step procedure of Section 5: moments are estimated by cell frequencies, and the structural system (5.1)–(5.2) is solved by a multistart Newton-type solver. Confidence intervals are constructed from 200 nonparametric bootstrap replications per Monte Carlo draw.

Table 1 reports bias, root mean squared error (RMSE), and empirical coverage of the nominal 95% confidence interval.

Table 1: Monte Carlo results: $n = 3{,}000$, $R = 300$ replications. True values: $\pi^1_{good} = 0.708$, $\pi^0_{good} = 0.544$, ATE $= 0.164$.

Estimand	**Truth**	**Bias**	**RMSE**	**Coverage**
$\mathbb{P}(Y_1 = good)$	0.708	-0.003	0.024	96.7%
$\mathbb{P}(Y_0 = good)$	0.544	-0.000	0.042	95.3%
ATE	0.164	-0.003	0.063	96.0%

The estimator is nearly unbiased across all three estimands, with bias below half a percentage point at $n = 3{,}000$. The slight undercoverage of π^0_{good} relative to π^1_{good} reflects the fact that identification of the untreated potential outcome distribution relies more heavily on the instrument through cross-equation moment restrictions and is therefore more sensitive to finite-sample noise in the first stage. Empirical coverage ranges from 95.3% to 96.7%, confirming that the bootstrap provides reliable inference at this sample size.

7. EMPIRICAL APPLICATION

7.1 Setting and Data

I implement the framework to study the effect of market access on occupational choice, using data from the Enquête sur les Conditions de Vie des Ménages (ECVM) for the Democratic Republic of Congo (DRC) 2024. The analysis focuses on how improvements in market access reshape the distribution of employment across broad economic sectors.

The unit of analysis is the household head. The outcome Y is a categorical indicator of the individual's main sector of activity, aggregated into three groups: (i) primary (agriculture, livestock, and fishing), (ii) industry (extractive activities, manufacturing, and construction), and (iii) services (commerce, transport, education, health, and other services). The treatment D is a binary indicator equal to one if the household's market access index exceeds the sample mean. The index aggregates ten commune-level indicators drawn from the ECVM commune survey module, spanning physical accessibility (five indicators measuring ease and reliability of transport) and service availability (five indicators measuring the presence of markets and commercial establishments). All indicators are recoded prior to PCA so that higher values correspond to better access. The instrument Z is a binary indicator of local topography (equal to one if the local terrain is predominantly plain). Topography affects the cost of building infrastructure and hence market access, but is plausibly exogenous to individual occupational choices conditional on controls.

Table 2 presents a summary of the data.

Table 2: Summary of data and variables.

Feature	**Value**
Observations	19,039
Outcome Y	Primary (51.7%), Industry (12.1%), Services (36.2%)
Treatment D	Above-mean market access (42.1% treated)
Instrument Z	Favorable topography (45.5% with $Z = 1$)
Covariates	Wealth index (continuous), gender, education level

The semiparametric two-step estimator is applied with covariates X entering both the selection equation and the outcome log-odds. Marginal potential outcome probabilities are obtained by averaging over the sample distribution of X. Inference relies on the nonparametric bootstrap. Table 3 reports the results.

Market access induces a substantial reallocation of labor across sectors. The probability of primary sector employment falls by 21.4 percentage points, from 64.6% without market access to 43.1% with it, and this effect is statistically significant at conventional levels. The probability of industrial employment rises by 13.6 percentage points, an increase that is also

Table 3: Estimated potential-outcome distributions and average treatment effects by sector. Bootstrap standard errors in parentheses; 95% confidence intervals in brackets.

Sector	$\hat{\pi}^1$	$\hat{\pi}^0$	$\widehat{\text{ATE}}$	**(S.E.)**	**95% CI**
Primary	0.431	0.646	−0.214	(0.091)	[−0.393, −0.036]
Industry	0.180	0.044	+0.136	(0.055)	[+0.028, +0.244]
Services	0.389	0.310	+0.079	(0.062)	[−0.043, +0.200]

significant. The effect on services is positive (+7.9 percentage points) but not statistically significant. The dominant adjustment margin is therefore a shift out of agriculture and into industry, consistent with the hypothesis that improved connectivity facilitates structural transformation by reducing the costs of accessing non-agricultural markets and employment opportunities.

These results illustrate the value of retaining the full distributional content of the categorical outcome. A binary IV analysis contrasting primary employment against all other activities would have captured only the first of these margins; the reallocation toward industry, which emerges clearly here, would have remained invisible.

8. CONCLUSION

This paper develops a nonparametric framework for identifying treatment effects when outcomes are categorical, extending the logic of instrumental variable methods beyond continuous and ordered settings. The starting observation is that researchers confronted with categorical outcomes routinely either discard distributional information by binarizing the outcome or impose restrictive parametric assumptions to recover it. This paper offers an alternative that retains full distributional content without parametric restrictions on the unobservables.

The central contribution is the association similarity assumption, which, combined with standard IV conditions and a local copula decomposition, yields point identification of the full distribution of potential outcomes for genuinely unordered categorical outcomes using only a binary instrument. The identification argument reduces to a one-dimensional

fixed-point problem per outcome category. Two partial identification results complement the main theorem by characterizing the identified set under monotonic association and bounded association, providing a transparent and formally sharp foundation for sensitivity analysis.

The semiparametric IV multinomial logit model translates the identification result into a tractable two-step estimator. Monte Carlo evidence confirms that the estimator is nearly unbiased and that bootstrap inference is reliable at sample sizes typical of survey data. The empirical application to the DRC illustrates the practical value: improved market access generates statistically significant shifts in the sectoral distribution of employment, a finding that would be obscured by collapsing the categorical outcome to a binary indicator.

Several extensions are left for future work. First, the framework could be extended to settings with a multivalued or continuous instrument. Second, sharp identification under partial monotonicity restrictions on the treatment response function, analogous to the no-defiers condition in binary IV models, merits investigation. Third, the sensitivity analysis under bounded association could be used to construct uniformly valid confidence sets robust to local violations of association similarity.

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SUPPLEMENTARY MATERIAL

Appendix A (Proofs). Proofs of all lemmas and propositions stated in the main text (see below).

Data and Replication Code. The replication dataset (`data_est.csv`, $n = 19{,}039$ observations) and R/Python code to replicate all estimation results and Monte Carlo simulations are available as supplementary files. The data file contains variables Y (categorical outcome, 3 levels), D (binary treatment), Z (binary instrument), and covariates X_1, X_2, X_3.

USE OF GENERATIVE AI

The author used generative AI (Claude Sonnet 4.6) for language editing, LaTeX formatting, and replication code generation. The author reviewed and is responsible for all final content.

REFERENCES

S. Acerenza. Testing instrument validity with multivalued treatments. *Working Paper*, 2023.

J. D. Angrist, G. W. Imbens, and D. B. Rubin. Identification of causal effects using instrumental variables. *Journal of the American Statistical Association*, 91:444–455, 1996.

A. Balke and J. Pearl. Bounds on treatment effects from studies with imperfect compliance. *Journal of the American Statistical Association*, 92:1171–1176, 1997.

A. Beresteanu and F. Molinari. Asymptotic properties for a class of partially identified models. *Econometrica*, 76:763–814, 2008.

V. Chernozhukov and C. Hansen. An IV model of quantile treatment effects. *Econometrica*, 73:245–261, 2005.

V. Chernozhukov, I. Fernández-Val, and A. E. Kowalski. Quantile regression with censoring and endogeneity. *Journal of Econometrics*, 186:201–221, 2015.

A. Chesher. Instrumental variable models for discrete outcomes. *Econometrica*, 78:575–601, 2010.

T. G. Conley, C. B. Hansen, and P. E. Rossi. Plausibly exogenous. *Review of Economics and Statistics*, 94:260–272, 2012.

J. A. Dubin and D. L. McFadden. An econometric analysis of residential electric appliance holdings and consumption. *Econometrica*, 52:345–362, 1984.

J. A. Dubin and D. Rivers. Selection bias in linear regression, logit and probit models. *Sociological Methods and Research*, 18:360–390, 1989.

D. A. Freedman and J. S. Sekhon. Endogeneity in probit response models. *Political Analysis*, 18:138–150, 2010.

S. Han. Identification in nonparametric models for dynamic treatment effects. *Journal of Econometrics*, 225:132–147, 2021.

S. Han and S. Lee. Estimation in a generalization of bivariate probit models with dummy endogenous regressors. *Journal of Applied Econometrics*, 34:994–1015, 2019.

J. L. Horowitz. Bootstrap methods in econometrics. *Annual Review of Economics*, 11: 193–224, 2019.

C. F. Manski. Nonparametric bounds on treatment effects. *American Economic Review*, 80:319–323, 1990.

M. Mogstad and A. Torgovitsky. Instrumental variables methods. *Annual Review of Economics*, 16:617–651, 2024.

I. Mourifie. Sharp bounds on treatment effects in a binary triangular system. *Journal of Econometrics*, 187:74–81, 2015.

J. R. Munkres. *Topology*. Prentice Hall, Upper Saddle River, NJ, 2nd ed. edition, 2000.

A. Nevo and A. M. Rosen. Identification with imperfect instruments. *Review of Economics and Statistics*, 94:659–671, 2012.

W. K. Newey. The asymptotic variance of semiparametric estimators. *Econometrica*, 62: 1349–1382, 1994.

P. R. Rosenbaum. Sensitivity analysis for certain permutation inferences in matched observational studies. *Biometrika*, 74:13–26, 1987.

A. M. Shaikh and E. J. Vytlacil. Partial identification in triangular systems of equations with binary dependent variables. *Econometrica*, 79:949–955, 2011.

W. P. M. M. van de Ven and B. M. S. van Praag. The demand for deductibles in private health insurance. *Journal of Econometrics*, 17:229–252, 1981.

E. Vytlacil and N. Yildiz. Dummy endogenous variables in weakly separable models. *Econometrica*, 75:757–779, 2007.

APPENDIX A: PROOFS

A.1 Proof of Lemma 3.1

Fix $z \in \{0,1\}$. By the switching equation $D = ZD_1 + (1-Z)D_0$, it follows that $D = D_z$ conditional on $Z = z$. Fix $k \in \{1, \ldots, q-1\}$. For each treatment state $d \in \{0,1\}$, define $\pi_k^d := \mathbb{P}(Y_d = c_k)$. Under Assumptions 1 and 2, the multinomial logit representation gives

$$\pi_k^d = \frac{e^{\mu_k^d}}{1 + \sum_{j=1}^{q-1} e^{\mu_j^d}} \in (0,1), \qquad d \in \{0,1\},$$

and $p_z := \mathbb{P}(D = 1 \mid Z = z) = \mathbb{P}(D_z = 1) \in (0,1)$.

Case $d = 1$. Define $m_{1,k,z} := \mathbb{P}(Y = c_k, D = 1 \mid Z = z)$. Using $D = D_z$ conditional on $Z = z$, together with the consistency relation $Y = Y_1$ on $\{D = 1\}$ and Assumption 1,

$$m_{1,k,z} = \mathbb{P}(Y_1 = c_k,\ D_z = 1).$$

Since $\mathbf{1}\{Y_1 = c_k\}$ and D_z are Bernoulli variables with means π_k^1 and p_z respectively, their joint probability satisfies the Fréchet–Hoeffding bounds:

$$\max\{0,\ \pi_k^1 + p_z - 1\} \ \le\ m_{1,k,z} \ \le\ \min\{\pi_k^1,\ p_z\}.$$

Under the interiority condition $|\omega_{d,k,z}| < 1$, the value $m_{1,k,z}$ lies in the strict interior of this interval. The map $\omega \mapsto \Lambda_2(u, v, \omega)$ is continuous and strictly increasing on $(-1, 1)$ for fixed $(u, v) \in (0,1)^2$, since $\partial_\omega \Lambda_2(u, v, \omega) = \phi_2(\Phi^{-1}(u), \Phi^{-1}(v); \omega) > 0$. Moreover, $\Lambda_2(u, v, \omega) \to \max\{u + v - 1, 0\}$ as $\omega \to -1$ and $\Lambda_2(u, v, \omega) \to \min\{u, v\}$ as $\omega \to 1$. By the Intermediate Value Theorem, there exists a unique $\omega_{1,k,z} \in (-1, 1)$ such that

$$m_{1,k,z} = \Lambda_2(\pi_k^1,\ p_z,\ \omega_{1,k,z}),$$

which is equation (3.1).

Case $d = 0$. Define $m_{0,k,z} := \mathbb{P}(Y = c_k, D = 0 \mid Z = z)$. Using $D = D_z$ and $Y = Y_0$ on $\{D = 0\}$,

$$m_{0,k,z} = \mathbb{P}(Y_0 = c_k,\ D_z = 0).$$

Here $\mathbf{1}\{Y_0 = c_k\}$ and $1 - D_z$ have means π_k^0 and $1 - p_z$, so by the same monotonicity argument there exists a unique $\tilde{\omega}_{0,k,z} \in (-1, 1)$ such that $m_{0,k,z} = \Lambda_2(\pi_k^0,\, 1 - p_z,\, \tilde{\omega}_{0,k,z})$. Setting $\omega_{0,k,z} := -\tilde{\omega}_{0,k,z}$ yields (3.2).

Sign relation. For any $(u, v) \in (0, 1)^2$, $\Lambda_2(u, v, \omega) \gtrless uv$ iff $\omega \gtrless 0$. Therefore,

$$\mathrm{Cov}\big(\mathbf{1}\{Y_d = c_k\},\, D_z\big) = \Lambda_2(\pi_k^d, p_z, \omega_{d,k,z}) - \pi_k^d\, p_z,$$

so $\mathrm{sign}(\omega_{d,k,z}) = \mathrm{sign}\big(\mathrm{Cov}(\mathbf{1}\{Y_d = c_k\}, D_z)\big)$. □

A.2 Proof of Proposition 3.2

The proof establishes global injectivity of the moment mapping $G_k : \mathcal{D}_k \to \mathbb{R}^4$, where $\mathcal{D}_k := (0, 1)^2 \times (-1, 1)^2$ with elements $(\pi^1, \pi^0, \omega_1, \omega_0)$, and

$$G_k(\pi^1, \pi^0, \omega_1, \omega_0) = \begin{pmatrix} \Lambda_2(\pi^1,\, p_1,\, \omega_1) \\ \Lambda_2(\pi^1,\, p_0,\, \omega_0) \\ \Lambda_2(\pi^0,\, 1 - p_1,\, -\omega_1) \\ \Lambda_2(\pi^0,\, 1 - p_0,\, -\omega_0) \end{pmatrix}.$$

I write the four observable moments as (a_1, a_0, b_1, b_0) and drop the category index k for readability.

Step 1: Decoupling across categories. Under Assumption 3, the four observable moments for category k depend only on the four parameters $(\pi^1, \pi^0, \omega_1, \omega_0)$ for that category. The full mapping is therefore block-diagonal with $(q - 1)$ independent blocks. It suffices to prove injectivity of each block.

Step 2: Implicit functions. Because Λ_2 is strictly increasing in each of its three arguments, each equation in G_k can be inverted uniquely.

- From $a_1 = \Lambda_2(\pi^1, p_1, \omega_1)$: for fixed π^1, there exists a unique $\omega_1 = \psi_1(\pi^1)$. Since increasing π^1 requires decreasing ω_1 to hold a_1 fixed, ψ_1 is strictly decreasing.
- From $b_1 = \Lambda_2(\pi^0, 1 - p_1, -\omega_1)$: for fixed ω_1, there exists a unique $\pi^0 = \varphi_1(\omega_1)$, which is strictly increasing.

- From $b_0 = \Lambda_2(\pi^0, 1 - p_0, -\omega_0)$: for fixed π^0, there exists a unique $\omega_0 = \chi_0(\pi^0)$, which is strictly increasing.

- From $a_0 = \Lambda_2(\pi^1, p_0, \omega_0)$: for fixed ω_0, there exists a unique $\pi^1 = \tau_0(\omega_0)$, which is strictly decreasing.

All four implicit functions are C^1 on their domains by the implicit function theorem and the smoothness of Λ_2.

Step 3: Reduction to a one-dimensional fixed point. Define $T : (0, 1) \to (0, 1)$ by

$$T(\pi^1) = \tau_0\big(\chi_0(\varphi_1(\psi_1(\pi^1)))\big).$$

A quadruple $(\pi^1, \pi^0, \omega_1, \omega_0)$ satisfies all four equations if and only if $\pi^1 = T(\pi^1)$, $\omega_1 = \psi_1(\pi^1)$, $\pi^0 = \varphi_1(\omega_1)$, $\omega_0 = \chi_0(\pi^0)$. Injectivity of G_k is therefore equivalent to uniqueness of the fixed point of T.

Step 4: Uniqueness via Assumption 4. Let $\delta_1(d, z)$ and $\delta_\omega(d, z)$ denote the partial derivatives of $\Lambda_2(\pi^d, p_z, \omega_z)$ with respect to its first and third arguments, and $\bar{\delta}_1(0, z)$, $\bar{\delta}_\omega(0, z)$ the corresponding derivatives for $\Lambda_2(\pi^0, 1-p_z, -\omega_z)$; all are strictly positive on $\mathcal{D}_k$. Define

$$\lambda(d, z) = \frac{\delta_\omega(d, z)}{\delta_1(d, z)} > 0, \qquad \bar{\lambda}(0, z) = \frac{\bar{\delta}_\omega(0, z)}{\bar{\delta}_1(0, z)} > 0.$$

By the implicit function theorem,

$$T'(\pi^1) = \frac{\bar{\lambda}(0, 1)}{\lambda(1, 0)\,\bar{\lambda}(0, 0)\,\lambda(1, 1)}.$$

Let I be the feasible set for π^1 (a connected open interval). Since T' is continuous on I and Assumption 4 implies $T'(\pi^1) \neq 1$ for all $\pi^1 \in I$, continuity and connectedness force either $T' < 1$ everywhere or $T' > 1$ everywhere on I.

- If $T' < 1$: T is a contraction; by Banach's fixed-point theorem, at most one fixed point exists.

- If $T' > 1$: the inverse T^{-1} satisfies $(T^{-1})' < 1$ and is a contraction; the equation $T(\pi^1) = \pi^1$ is equivalent to $T^{-1}(\pi^1) = \pi^1$, which has at most one solution.

In either case T has at most one fixed point on I, so G_k is injective and the true parameter is uniquely determined. $\square$

A.3 Proof of Proposition 4.1

Under Assumption 5, for each k and z, $\omega_{1,k,z} \geq \omega_{0,k,z}$. Any tuple $(\pi_k^0, \pi_k^1, \omega_{0,k,z}, \omega_{1,k,z}) \in (0,1)^2 \times (-1,1)^2$ satisfying

$$\begin{aligned} m_{k,z}^{(1)} &= \Lambda_2(\pi_k^1,\, p_z,\, \omega_{1,k,z}), \\ m_{k,z}^{(0)} &= \Lambda_2(\pi_k^0,\, 1-p_z,\, -\omega_{0,k,z}), \\ \omega_{1,k,z} &\geq \omega_{0,k,z} \end{aligned}$$

is observationally equivalent to the true parameter. The identified set $\mathcal{B}_k^{\text{mono}}$ is the projection of this feasible set onto (π_k^0, π_k^1).

Sharpness. Every point in $\mathcal{B}_k^{\text{mono}}$ is achieved by choosing appropriate $(\omega_{0,k,z}, \omega_{1,k,z})$ with $\omega_{1,k,z} \geq \omega_{0,k,z}$, because for any fixed $(\pi_k^0, \pi_k^1) \in (0,1)^2$ the moment equations uniquely determine $\omega_{d,k,z}$ as a function of π_k^d. The constraint $\omega_{1,k,z} \geq \omega_{0,k,z}$ defines a half-space in the (ω_0, ω_1) plane that is attained.

Nesting. When $\omega_{1,k,z} = \omega_{0,k,z}$, the set collapses to the point-identified value of Proposition 3.2. The identified set strictly contains this singleton whenever the inequality is not binding. $\square$

A.4 Proof of Proposition 4.2

Under Assumption 6 with sensitivity parameter $\delta \geq 0$, the feasible set is

$$\begin{aligned} \mathcal{F}_k^\delta = \big\{ &(\pi_k^0, \pi_k^1, \omega_{0,k,z}, \omega_{1,k,z}) \in (0,1)^2 \times (-1,1)^2 : \\ &\qquad \text{equations (3.1)–(3.2) hold, } |\omega_{1,k,z} - \omega_{0,k,z}| \leq \delta \big\}, \end{aligned}$$

and $\mathcal{B}_k^\delta$ is its projection onto (π_k^0, π_k^1).

Part (i). When $\delta = 0$ the constraint reduces to Assumption 3, and Proposition 3.2 gives a singleton.

Part (ii). For $\delta' > \delta \geq 0$, $\mathcal{F}_k^\delta \subseteq \mathcal{F}_k^{\delta'}$, so $\mathcal{B}_k^\delta \subseteq \mathcal{B}_k^{\delta'}$. Compactness of $\mathcal{B}_k^\delta$ follows from $\mathcal{F}_k^\delta$ being a closed and bounded subset of a compact set and projection being continuous.

Connectedness follows from the convexity of the constraint $|\omega_{1,k,z} - \omega_{0,k,z}| \leq \delta$ and the continuity of G_k (Munkres 2000): the image of a connected set under a continuous map is connected.

Part (iii). As $\delta \to \infty$, the constraint becomes vacuous. The only binding restrictions are the moment equations and the requirement $(\omega_{0,k,z}, \omega_{1,k,z}) \in (-1,1)^2$. For any fixed $(\pi_k^0, \pi_k^1) \in (0,1)^2$ the moment equations can be satisfied for some $(\omega_{0,k,z}, \omega_{1,k,z}) \in (-1,1)^2$ whenever $m_{k,z}^{(d)}$ lies in the strict interior of the Fréchet–Hoeffding bounds, which holds under the maintained assumptions. The limiting identified set therefore coincides with the Manski bounds [Manski, 1990]. □